SUBOBJECTS OF THE FINAL OBJECT AND SUBTOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The slice topos and its localization morphism. Same chapter, Lemma lemma-localize-topos. Supplies the identification of the slice of the topos with the localized category of sheaves.
- Replacing the topos by a good site. Same chapter, Lemma lemma-topos-good-site. Allows the author to assume that the sheaf is representable by an object of a site with subcanonical topology and finite limits.
- The pushforward along a localization when the site has products. Same chapter, Lemma lemma-localize-given-products and Lemma lemma-restrict-back. Supply the computations of the pushforward and of the composite identity used in the two directions.

Source reference. Original lemma 04GZ: sources/stacks-sites/pinned-original-source.tex, source lines 10343–10343. Navigation pointer invoked at proof line 10343; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CI: sources/stacks-sites/pinned-original-source.tex, source lines 10345–10345. Navigation pointer invoked at proof line 10345; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03CE: sources/stacks-sites/pinned-original-source.tex, source lines 10362–10362. Navigation pointer invoked at proof line 10362; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 001P: sources/stacks-sites/pinned-original-source.tex, source lines 10368–10368. Navigation pointer invoked at proof line 10368; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 08LR: sources/stacks-sites/pinned-original-source.tex, source lines 10370–10370. Navigation pointer invoked at proof line 10370; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00Y2: sources/stacks-sites/pinned-original-source.tex, source lines 10375–10375. Navigation pointer invoked at proof line 10375; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{F}$ be a sheaf on $\mathcal{C}$. The following are equivalent*

1. *$\mathcal{F}$ is a subobject of the final object of $Sh(\mathcal{C})$, and*
2. *the topos $Sh(\mathcal{C})/\mathcal{F}$ is a subtopos of $Sh(\mathcal{C})$.*

Proof. We have seen in Lemma 04GZ that $Sh(\mathcal{C})/\mathcal{F}$ is a topos. In fact, we recall the proof. First we apply Lemma 03CI to see that we may assume $\mathcal{C}$ is a site with a subcanonical topology, fibre products, a final object X, and an object U with $\mathcal{F} = h_U$. The proof of Lemma 04GZ shows that the morphism of topoi $j_{\mathcal{F}} : Sh(\mathcal{C})/\mathcal{F} \to Sh(\mathcal{C})$ is equal (modulo certain identifications) to the localization morphism $j_U : Sh(\mathcal{C}/U) \to Sh(\mathcal{C})$.

Assume (2). This means that $j_U^{-1} j_{U,*}\mathcal{G} \to \mathcal{G}$ is an isomorphism for all sheaves $\mathcal{G}$ on $\mathcal{C}/U$. For any object Z/U of $\mathcal{C}/U$ we have

$$(j_{U,*}h_{Z/U})(U) = \mathrm{Mor}_{\mathcal{C}/U}(U \times_X U/U, Z/U)$$

by Lemma 03CE. Setting $\mathcal{G} = h_{Z/U}$ in the equality above we obtain

$$\mathrm{Mor}_{\mathcal{C}/U}(U \times_X U/U, Z/U) = \mathrm{Mor}_{\mathcal{C}/U}(U, Z/U)$$

for all Z/U. By Yoneda's lemma (Categories, Lemma 001P) this implies $U \times_X U = U$. By Categories, Lemma 08LR $U \to X$ is a monomorphism, in other words (1) holds.

Assume (1). Then $j_U^{-1} j_{U,*} = \mathrm{id}$ by Lemma 00Y2. □